CHENGKANG Zhou

The University of Hong Kong (HKU) - Hong Kong

(852) 52297695

zhouchk2@hku.hk

Personal Website

June 15, 2026
The City University of Hong Kong (CityUHK)
KOWLOON, HONG KONG

RE : Application for the academic position in Physics

Dear Members of the Faculty Search Committee,
Department of Physics, City University of Hong Kong,

It is with great enthusiasm that I submit my application for the academic position in the Department of Physics at City University of Hong Kong (CityUHK). I am currently a Postdoctoral Fellow at The University of Hong Kong, where I also earned my Ph.D. in Physics. My research focuses on the quantum magnetism, dynamical spectra, and large-scale numerical methods for quantum many-body systems. I have a strong track record of publishing in high-impact journals and collaborating with experimental groups, including Nat. Phys. 20, 1097–1102 (2024), Nat. Phys. 20, 1764–1771 (2024), and arXiv:2510.14813 (2025), which has been accepted by Nature Communications. Should you wish to learn more, please visit my personal website for additional information.

My recent research focuses on collective excitations and entanglement in frustrated quantum systems. Specifically, I have investigated the entanglement properties of quantum spin liquid (QSL) states on the pyrochlore lattice using quantum Monte Carlo (QMC) simulations, by extracting the quantum Fisher information (QFI) from dynamical structure factors [arXiv:2510.14813 (2025)]. QFI, which reflects how massively entangled a given state is, serves as both a promising and experimentally accessible probe for detecting the emergence of the quantum spin liquid phase. However, performing highly efficient QMC simulations on this model has been a long-standing challenge, due to the highly frustrated nature of the pyrochlore lattice and the ultra-low temperatures required to enter the QSL regime. To overcome this challenge, I have developed advanced numerical methods, including a multi-directed loop update scheme. With this methodological improvement, we have mapped out the thermal and quantum phase diagram from QFI measurements, which reveal how the entanglement structure changes as the system enters the QSL regime. This study serves as an important bridge, connecting the theoretical understanding of entanglement properties to experimentally accessible observations, and was recently accepted by *Nat. Commun.*. Building on this progress, I have extended this framework to study unconventional quantum critical points in kagome quantum spin liquids [arXiv:2603.19951 (2026)].

Besides, I have extensive experience in studying spin spectral functions and phase transitions in novel quantum spin systems, ranging from the Higgs mode in spin spectral functions to fracton topological order. I have applied a variety of methods, from numerical techniques such as quantum Monte Carlo to analytical theoretical approaches including spin wave theory and the Schwinger boson picture. My previous work has been published in journals such as Physical Review Letters, Nature Physics, and Nature Communications, among others. I believe that my research experience aligns well with the research directions at City University of Hong Kong, particularly in the areas of Spectroscopy and Imaging, as well as Theoretical and Computational Physics. This has motivated me to write this letter and submit my application. Beyond my research, I have participated in other academic activities. For example, I helped organize a conference on Fractional Chern Insulators, taking on the role of finance assistant. Recently, I was honored to be invited as a speaker at a conference in Cergy, France (conference link), where I shared my recent work on quantum Fisher information.

If given this opportunity, I believe my future research plans align well with the strategic directions of CityUHK. I would hope to apply for funding from the Research Grants Council (RGC) of Hong Kong, such as the General Research Fund (GRF) and the Early Career Scheme (ECS), to support my research activities. My future research plans can be broadly divided into two main directions.

First, I would like to further explore the relationship between spectral properties and entanglement structures in quantum spin systems. The entanglement properties of quantum many-body systems, such as entanglement entropy, are fundamental to understanding their quantum phases and phase transitions. However, directly measuring these properties in experiments is often challenging. Recently, a promising bridge has emerged from the combination of quantum information theory and many-body physics: QFI, which helps connect theoretical understanding of entanglement properties with experimentally accessible observations. My recent work has made some progress in this direction, but much remains to be explored. In particular, how QFI can be used to extract more detailed entanglement information from spin spectral functions is still an open question and attracts considerable interest. Therefore, if given the opportunity, I would like to build on my previous work on QFI and aim to develop a framework that extracts entanglement properties from spectral functions. A promising path forward would be to introduce additional tools from quantum information theory, such as the quantum Cramér–Rao bound, to further explore QFI and its relationship with entanglement structures in quantum spin systems. I believe this direction would not only deepen our understanding of the link between spectral properties and entanglement in quantum spin systems but also provide a useful tool for analyzing experimental data and identifying novel quantum phases.

Second, I hope to continue my research on quantum spin liquids, with a particular focus on kagome lattice systems such as $YCu_3(OD)_6Br_2[Br_x(OD)_{1-x}]$ in Nat. Phys. 20, 1097–1102 (2024). This material shows great promise for realizing quantum spin liquid states, as evidenced by the absence of magnetic order, the specific heat behavior, and the spin spectral function observed in inelastic neutron scattering experiments. However, the theoretical understanding of this material is still in its early stages, and there remain many open questions regarding the nature of its ground state and excitations. Theoretically, the kagome lattice in this material is believed to be a distorted kagome lattice. Compared to the conventional kagome lattice, the distorted version has a lower symmetry and a much larger unit cell, which makes numerical simulation and theoretical analysis more challenging. Nevertheless, recent advances in theoretical and numerical methods, such as Neural Quantum States [arXiv:2402.09402 (2024)] and Schwinger boson mean-field theory [Phys. Rev. B 98, 184403 (2018)], have made it possible to study such complex systems. Motivated by these observations and developments, I would like to continue my research on possible quantum spin liquids on the distorted kagome lattice. This direction would not only help reveal the nature of this material but also aim to develop a general theoretical framework for Schwinger boson calculations and to apply Neural Quantum States to other novel quantum systems.

I believe CityUHK's growing reputation in condensed matter theory and quantum materials, combined with its supportive environment for early-career faculty, would make it an ideal place for me to establish my research program. I would be truly excited about the opportunity to contribute to the Department's research excellence, to teach at both the undergraduate and graduate levels, and to mentor students from diverse backgrounds in such a vibrant, world-class city.

Thank you very much for considering my application. I would be most grateful for the chance to discuss how my expertise and vision might help support the continued success of the Department of Physics at CityUHK. Should you need any further information from me, please feel free to let me know and I would be very happy to provide it

Sincerely,
CHENGKANG Zhou
Chengkang zhou